ENTERPRISE ARCHITECTURE & SOLUTION DESIGN

Project Management Solution

Comprehensive Technical Solution Design Document (SDD) for Salesforce Core

PLATFORM	VERSION	ARCHITECTURE TYPE	ARTIFACT FORMAT
Salesforce Lightning	1.0 (Initial Release)	Declarative / Low-Code	Unmanaged Package & Metadata

1. Executive Summary & Business Need

Modern service and delivery organizations require structured, standardized project execution integrated directly with their core CRM accounts. The **Project Management Solution** bridges the gap between account-level account management and post-sale operational delivery by introducing automated project onboarding, rigid data governance, streamlined UX workflows, and real-time executive visibility.

Core Business Objectives

- **Data Governance:** Eliminate orphaned records and enforce accountability through relational linkage to Account records.
- **Operational Efficiency:** Reduce onboarding friction by providing a 3-field wizard directly from customer accounts.
- **Automated Execution:** Ensure instant task assignments upon project kickoff without manual admin overhead.
- **Executive Visibility:** Supply leadership with real-time portfolio dashboards tracking workload distribution.

2. Data Model & Architecture (ERD)

The data model introduces an independent Custom Object (`Project__c`) designed with standard enterprise conventions, configured with master-detail style accountability via a relational Lookup to standard CRM accounts.

Field Label	API Name	Data Type	Business Logic / Constraints
Project Name	`Name`	Text (80)	Standard identifier, required on creation.
Account	`Account__c`	Lookup(Account)	Relational parent linking each project to its billing/servicing account.
Status	`Status__c`	Picklist	Values: New (Default), `In Progress`, `Completed`, `On Hold`.
End Date	`End_Date__c`	Date	Represents actual completion date. Mandatory on project completion.
Project Owner	`OwnerId`	Lookup(User)	Inherited ownership driving role hierarchy and automated task delegation.

3. Business Logic & Data Governance

To ensure reporting accuracy and prevent projects from being marked complete without project closure audits, an automated validation rule enforces data completeness.

Validation Rule: Require End Date When Completed

- **API Name:** `Require_End_Date_When_Completed`

- **Object:** `Project__c`
- **Error Message:** *"An End Date is required before marking the project as Completed."*
- **Formula Definition:**

```
AND(
    ISPICKVAL(Status__c, "Completed"),
    ISBLANK(End_Date__c)
)
```

4. Process Automation & Workflow Engineering

The automation strategy follows Salesforce best practices: declarative flows handle asynchronous backend triggering and synchronous guided user intake.

Automation Name	Type	Trigger / Invocation	Core Architectural Responsibilities
Project - Quick Create Wizard	Screen Flow	Account Quick Action button	• Receives `recordId` context variable dynamically from current Account. • Displays user-friendly modal requesting Name and Status. • Executes `Create Records` mapping Account ID directly without manual data entry.
Project - Create Kickoff Task	Record-Triggered Flow	`Project__c` Updated: `Status__c` equals `In Progress`	• Evaluates formula condition: Only when a record is updated to meet the condition requirements. • Instantiates standard `Task` record linked via `WhatId = Project.Id`. • Assigns task to `OwnerId` with priority "High" and due date calculated 7 days out.

5. UI/UX & Lightning Experience Customization

Adoption was optimized by minimizing administrative clicks and organizing the Lightning Record Page according to operational cognitive flow:

- **Header Area:** Highlights critical project status, Account lookup, and End Date using Compact Layout.
- **Split Pane Layout:** Left panel reserved for structured record fields; right panel dedicated to the **Activity Timeline** showing automated Kickoff tasks, emails, and follow-ups.
- **Account Action Integration:** The Quick Action *"New Project"* is embedded into the Account layout, eliminating context-switching between tabs.

6. Analytics & Executive Business Intelligence

Reporting is configured natively with public folder governance, enabling organizational visibility while preventing unfiled metadata degradation.

Analytics Asset	Folder Container	Configuration & Metrics
Projects by Status (Report)	`Project Reports` (Custom Public)	• Type: Summary Report. • Group Rows: Grouped logically by `Status__c`. • Columns: `Project Name`, `Account`, `End Date`, `Project Owner`.

Analytics Asset	Folder Container	Configuration & Metrics
Executive Project Overview (Dashboard)	`Project Dashboards` (Custom Public)	• **Widget 1 (Donut Chart):** Breakdown of active projects categorized across lifecycle stages. • **Widget 2 (Metric Scorecard):** Total Active Projects KPI metric count.

7. Release, Packaging & Deployment Specifications

The complete solution has been decoupled from local developer org dependencies and bundled into an enterprise-ready installation package.

Deployment Artifact Summary

Package Name: Project Management Solution | **Version:** 1.0 (Initial Release)

Installation URL:

```
https://login.salesforce.com/packaging/installPackage.apexp?p0=04tgK000000Js4j
```

Extracted Metadata Package Components:

- `objects/Project__c.object` (Custom Object, Fields, and Validation Rules)
- `flows/Project_Quick_Create_Wizard.flow` & `flows/Project_Create_Kickoff_Task.flow`
- `flexipages/Project_Record_Page.flexipage` & `layouts/Project__c-Project Layout.layout`
- `dashboards/Project_Dashboards/Executive_Project_Overview.dashboard`
- `reports/Project_Reports/Projects_by_Status.report`
- `package.xml` (Project Manifest)